Shubham Jain

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Education

- **Purdue University - West Lafayette** *Aug 2023 – Dec 2026*
  B.S. in Computer Science, B.S. in Artificial Intelligence GPA: 3.80

Technical Skills

- **Languages**: Python, Java, C/C++, Rust, SQL, R, HTML/CSS/JS.
- **AI/ML Frameworks**: TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, PySpark.
- **Cloud & Infra**: AWS, Azure, Docker, Databricks (Lakehouse), Firebase.
- **Web & Systems**: React.js, Node.js, Salesforce (Apex, LWCs, Data Cloud, Agentforce), Databricks, Firebase.
- **DevOps & Tools**: Git/GitHub, CI/CD (GitHub Actions), Linux/Unix.

Experience

- **Amazon Web Services** *May 2026 – Aug 2026*
  SDE Intern - AWS Quick Work Seattle, WA
  - Engineered an end-to-end conversation recap feature for Amazon Quick, leveraging on-demand summaries to cut user context-recovery time **by 2+ minutes** to a few seconds with **0 idle compute overhead**.
  - Built a modular fidelity testing framework in Python comparing live APIs (Outlook, Slack, Gmail) against digital twins across **70+ API actions** to **improve simulation accuracy by 32%**.
  - Automated multi-target test loops enabling developers to rapidly onboard new connectors in Rust and execute comparative validation pipelines, reducing effort **from days to hours.**

- **Robert Half / Protiviti** *Jun 2025 – May 2026*
  Software Engineering Intern San Ramon, CA
  - Built an ML pipeline in Python and Salesforce Data Cloud, improving prediction accuracy from **72%** to **86%**.
  - Shipped **20+ LWCs** and **40+ Apex classes** for **30+ production tickets** to enhance functionality for consultants.
  - Engineered **4 autonomous AI agents** via **Agentforce** to automate record creation and edits, reducing effort by **60%**.

- **Indiana Fever (with The Data Mine)** *Aug 2024 – May 2025*
  Machine Learning Lead: Data Science Researcher Indianapolis, IN
  - Led development of supervised ML models (**XGBoost, RF**) predicting WNBA player performance with **80% accuracy**.
  - Cut model **RMSE by 23%** via hyperparameter tuning, ensembling, and feature importance analysis.
  - Architected a Databricks pipeline to clean, aggregate, and transform **500K+ player stats** for downstream analytics.

- **Elastik Teams** *Jun 2022 – Aug 2022*
  Software Engineering Intern Remote
  - Built RESTful APIs in Node/Express with **OAuth 2.0**, cutting login latency by **20%** with **98% coverage** in **30+ tests**.
  - Redesigned microservices architecture and **CI/CD pipelines**, reducing deployment time from **1 hr to 10 mins**.

Projects

- **Slingshot Sumo** | Node.js, Express.js, Socket.IO, Three.js, Matter.js
  Real-time multiplayer 3D physics game with lobby system and cross-platform support
  - Architected **real-time multi-client web game** using Socket.IO with lobby management, invite links, and host controls.
  - Integrated **Matter.js** physics engine for collision detection and trajectory simulation on a dynamically shrinking arena.
  - Rendered toggleable **3D/2D** views with Three.js; implemented responsive touch controls for full mobile support.

- **Compiler** | C, Assembly Language, Lex, Yacc, Linux/Unix, Compiler Design, Systems Programming
  SimpleC compiler with lexical analysis, parsing, and code generation
  - Developed a compiler with lexical analysis, parsing, and **x86_64 assembly** code generation.
  - Implemented semantic analysis for type checking, scope resolution, and compile-time error detection.
  - Optimized **AST-to-assembly translation** for improved performance and correctness.

- **Sort-o-Matic** | Python, TensorFlow, OpenCV, Firebase, JavaScript, HTML/CSS
  Computer vision waste classification system with CNN and IoT integration
  - Trained a CNN in TensorFlow to classify waste in real-time with **85% accuracy**.
  - Built an IoT-enabled dashboard (Firebase backend + web UI) for device control and analytics.
  - Awarded **3rd place** among 600+ participants at the EcoHacks hackathon.